EXPERIMENT 5

IPC USING SHARED MEMORY

AIM: To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

SOFTWARE REQUIRED: Oracle VirtualBox, Fedora Linux, GCC compiler, Terminal.

ALGORITHM:

SENDER:

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Write a string to the shared memory segment using sprintf
5. Set delay using sleep
6. Detach shared memory segment using shmdt

RECIEVER:

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Print the shared memory contents sent by the sender

process.

5. Detach shared memory segment using shmdt

PROGRAM:

(A) SENDER.C

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/ipc.h>
#include <sys/shm.h>
```

```
int main()
{
    int shmid;
    char *str;

    shmid = shmget(1234, 1024, 0666 | IPC_CREAT);

    if (shmid == -1)
    {
        perror("shmget");
        return 1;
    }

    str = (char *)shmat(shmid, NULL, 0);

    if (str == (char *)-1)
    {
        perror("shmat");
        return 1;
    }

    printf("Enter the message: ");
    fgets(str, 1024, stdin);

    printf("Message sent: %s", str);

    shmdt(str);

    return 0;
}
```

(B) RECEIVER.C

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main()
{
    int shmid;
    char *str;

    shmid = shmget(1234, 1024, 0666);

    if (shmid == -1)
    {
        perror("shmget");
        return 1;
    }
```

```
    str = (char *)shmat(shmid, NULL, 0);

    if (str == (char *)-1)
    {
        perror("shmat");
        return 1;
    }

    printf("Message received: %s", str);

    shmdt(str);
    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

OUTPUT:



RESULT:

Thus, Inter-Process Communication using shared memory was implemented and executed successfully in the Fedora Linux operating system.